Work through the following exercises while the TA circulates. When you have completed the exercises, spend the rest of the laboratory working on the homework assignment.

Warm up.

Compute the derivative of the following expressions.

1. $3e^x$.
2. $3xe^x$.
3. $x\sin(x)$.
4. $\frac{x\cos(x)}{x^2}$.
5. $e^2e^xx^{-1}$ using the product rule.
6. $\frac{e^2e^x}{x}$ using the quotient rule.

Exercise 1.

Consider the function

$$f(x) = \frac{x^2 + 4x + 4}{x + 2}.$$

1. Find the equation $L(x) = mx + b$ of the line tangent to $f(x)$ at the point $(2, 4)$.
2. Find the equation $N(x) = nx + c$ of the line perpendicular to $L(x)$ at the point $(2, 4)$. This line is called the *normal line* of $f(x)$ at the point $(2, 4)$.

Exercise 2.

Recall that a line can be given as a function $L(x) = mx + b$, where m is the slope of the line and b is given by the point where the line intersects the y-axis. Using your answer from the previous exercise, find the equation of the line tangent to the following functions at the following points.

1. Compute the derivative of $x(x + 1)^2$ using the product rule.
2. Expand the expression $x(x + 1)^2$ and compute its derivative by using the power rule.
3. Explain the relation between these two derivatives.